MD MAINUL ISLAM

ACADEMIC CV | PhD Applicant

Dhaka, Bangladesh • mainul.ptfaculty@cse.uiu.ac.bd • linkedin.com/in/mainulfuad

RESEARCH PROFILE

Computer Science and Engineering graduate (CGPA 3.92/4.00, Summa Cum Laude) with undergraduate research experience in AI-enabled assistive technology for dementia care, first-person video activity recognition, deep learning, and epidemiological simulation. Published work in software engineering and AI in healthcare. Research interests include trustworthy and explainable AI, healthcare AI, computer vision, multimodal learning, predictive analytics, and reliable machine learning systems.

EDUCATION

BSc in Computer Science and Engineering | United International University (UIU), Bangladesh

Completed Summer 2020 • Degree awarded February 2024 • 138 credits • CGPA 3.92/4.00

Summa Cum Laude • Best Thesis Award • Seven 100% merit scholarships based on academic performance

Completed a two-part Final Year Design Project (CSI 4000A and CSI 4000B). English was the medium of instruction.

RESEARCH EXPERIENCE

The Third-Eye: A Video-Based Helping Aid for Dementia Patients | Final Year Design Project, UIU

• Designed a video-based assistive system concept to recognize activities of daily living from a first-person camera view and support reminders when routine activities were not detected within expected time windows.

• Captured activity video using a Jetson Nano and night-vision camera; processed video at 8 FPS, downsampled to 4 FPS, and organized five-second temporal sequences.

• Used MobileNetV2 for frame-level visual feature extraction and LSTM for temporal activity classification, comparing the LSTM model with a simple RNN.

• The project report records 91.67% LSTM accuracy, 81.22% specificity, 79.81% sensitivity, and 80.03% F1 score, compared with 25.26% RNN accuracy. The reminder component was work in progress, not a clinically validated intervention.

• Evaluated 13 activity classes, including hand washing, taking medicine, drinking water, reading, and brushing teeth.

Effects of Social Distancing During Coronavirus in Bangladesh | Simulation Modeling Laboratory, UIU (2020)

• Implemented a SEIRD differential-equation model to explore COVID-19 infection and mortality trajectories in Bangladesh and examine hypothetical social-distancing scenarios.

• Preprocessed historical case data, explored parameter adjustments including an assumed exposed population and a social-distancing parameter (RHO), and compared short-horizon predictions with reported infection and death counts.

• The project report acknowledged limitations in forecasting accuracy and data availability.

PUBLICATIONS

Islam, M. M. (2026). “Whose Specification Did You Test? The Hidden Interpretation Bias in LLM-Generated Test Suites.” RASE 2026, co-located with ASE 2026. ACM Digital Library. https://doi.org/10.1145/3820756.3844908

Islam, M. M. (2026). “Artificial Intelligence in Healthcare: Predictive Analytics for Early Detection of Chronic Diseases.” International Journal of Medical and Health Research. https://doi.org/10.61424/ijmhr.v4i1.852

ACADEMIC AND PROFESSIONAL EXPERIENCE

Adjunct Faculty (Industry), Department of CSE | United International University | 2025–Present

• Teach System Analysis and Design Laboratory; guide student teams on requirements analysis, system design, prototyping, documentation, and project presentations.

Senior Software Quality Assurance Engineer | ADN Telecom Limited (assigned to BRAC Bank IT) | October 2026–Present

• Conduct API testing and validation for banking application services, including service-level checks and database validation involving Oracle and MongoDB in a microservice-based environment.

Project Manager | WebAlive | May 2023–September 2026

• Coordinated software delivery, requirements, cross-functional execution, and product quality.

Software Engineer, Testing | Bit Mascot (Pvt.) Ltd. | September 2020–April 2023

• Performed software testing, test execution, defect reporting, and quality assurance.

TECHNICAL AND RESEARCH SKILLS

AI and research: Computer vision, MobileNetV2, LSTM, RNN, SEIRD modelling, predictive analytics, video processing, experimental evaluation.

Software quality and engineering: API testing, Postman, Playwright, Selenium, Cypress, JMeter, k6, Python, SQL, database testing, Jira, Confluence, GitHub.

AI-assisted engineering: Claude, Claude Code, Playwright MCP, AI-assisted test generation and evaluation.

HONOURS AND AWARDS

• Summa Cum Laude, BSc in Computer Science and Engineering, United International University.

• Best Thesis Award, United International University.

• Seven 100% merit scholarships awarded based on academic performance.

ENGLISH PROFICIENCY

IELTS Academic: Overall band score 6.0. • Medium of Instruction (MOI) certificate available.

ACADEMIC REFEREE

Professor Dr. Swakkhar Shatabda | BRAC University, Bangladesh

Undergraduate project supervisor and academic referee. Recommendation letter available upon request.

RESEARCH AND PROFESSIONAL LINKS

Research profile: https://researchbymainul.com/

Portfolio: https://mainulislam-portfolio.vercel.app/

LinkedIn: https://www.linkedin.com/in/mainulfuad